PSEUDO-COHERENCE OF DERIVED GLOBAL SECTIONS TESTED ON PERFECT COMPLEXES EXTENDS TO ALL PSEUDO-COHERENT COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a quasi-compact quasi-separated scheme the global sections functor on quasi-coherent complexes has finite cohomological dimension. used at proof lines 8874-8877. Fixes the integer N that controls how far up the cone can contribute after taking sections.
- Any pseudo-coherent complex on a quasi-compact quasi-separated scheme admits an approximation by a perfect complex up to any prescribed degree. invoked at proof lines 8882-8886. Provides the perfect complex P and the cone C whose low cohomology sheaves agree with E.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Cited at proof lines 8874-8877 to fix N; the degree bookkeeping consuming N is inside the counted span.

Source reference. Original theorem 08ES: sources/stacks-perfect/pinned-original-source.tex, source lines 4031–4031. Cited at proof lines 8883-8886 to obtain the approximation P of E; the triangle construction and degree bookkeeping are inside the counted span.

Lemma 0.2. *Let A be a ring. Let X be a scheme over A which is quasi-compact and quasi-separated. Let $K \in D^-_{QCoh}(\mathcal{O}_X)$. If $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every perfect E in $D(\mathcal{O}_X)$, then $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every pseudo-coherent E in $D(\mathcal{O}_X)$.*

Proof. There exists an integer N such that $R\Gamma(X, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ has cohomological dimension N as explained in Lemma 08D5. Let $b \in \mathbf{Z}$ be such that $H^i(K) = 0$ for $i > b$. Let E be pseudo-coherent on X. It suffices to show that $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is m-pseudo-coherent for every m. Choose an approximation $P \to E$ by a perfect complex P of $(X, E, m - N - 1 - b)$. This is possible by Theorem 08ES. Choose a distinguished triangle

$$P \to E \to C \to P[1]$$

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in $D_{QCoh}(\mathcal{O}_X)$. The cohomology sheaves of C are zero in degrees $\geq m - N - 1 - b$. Hence the cohomology sheaves of $C \otimes^{\mathbf{L}} K$ are zero in degrees $\geq m - N - 1$. Thus the cohomology of $R\Gamma(X, C \otimes^{\mathbf{L}} K)$ are zero in degrees $\geq m - 1$. Hence

$$R\Gamma(X, P \otimes^{\mathbf{L}} K) \to R\Gamma(X, E \otimes^{\mathbf{L}} K)$$

is an isomorphism on cohomology in degrees $\geq m$. By assumption the source is pseudo-coherent. We conclude that $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is m-pseudo-coherent as desired. $\square$